Interactive Urban Heat Island Explorer: A Web GIS for Green Space Accessibility and Local Heat Analysis

Student: Maria Mushtaq

Overview:

I will develop a browser-based Web GIS application that visualizes urban heat patterns together with green spaces and buildings. Users can click anywhere on the map to perform real-time spatial analysis and explore how nearby vegetation relates to local heat conditions.

Raster Datasets:

It might change the sources of the datasets in process of development of project.

- **Dataset:** Land Surface Temperature (LST) raster (or a heat map derived from satellite imagery)
- **Source:** Sentinel-3 LST or Landsat 8/9 thermal data processed through Google Earth Engine or Copernicus/Open Data sources.

Vector Dataset:

It might change the sources of the datasets in process of development of project.

- **Parks and Green Spaces:** OpenStreetMap (GeoJSON)
- **Buildings:** OpenStreetMap building footprints (GeoJSON)
- **Optional Trees or Vegetation Points:** OpenStreetMap or local open data portals

Spatial Computation:

When a user clicks on the map, the application will:

- Create a 500-meter buffer around the selected location using Turf.js.
- Count the number of parks located within the buffer.
- Calculate the distance to the nearest green space.
- Determine the average land surface temperature within the selected area (or report the local raster value).
- Display all computed statistics in an interactive popup or side panel.

Tool:

- Map Libre GL JS – interactive Web GIS visualization
- Turf.js – client-side spatial computations
- HTML, CSS, JavaScript – frontend development
- GeoJSON – vector data format

- GeoTIFF or raster tiles – heat map visualization

Expected Outcome:

Urban heat islands are an important environmental challenge in modern cities. This project combines raster and vector geospatial data with real-time client-side spatial analysis to demonstrate how Web GIS can support environmental awareness and smart urban planning.